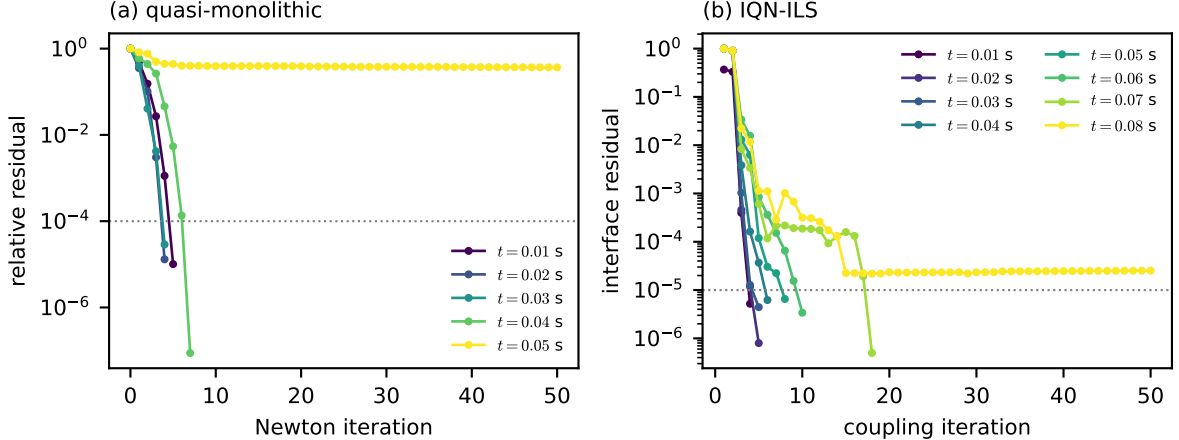
(a) quasi-monolithic
relative residual
10^{0}
10^{-2}
10^{-4}
10^{-6}
0
10
20
30
40
50
Newton iteration
$t = 0.01$ s
$t = 0.02$ s
$t = 0.03$ s
$t = 0.04$ s
$t = 0.05$ s
(b) IQN-ILS
interface residual
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
10
20
30
40
50
coupling iteration
$t = 0.01$ s
$t = 0.02$ s
$t = 0.03$ s
$t = 0.04$ s
$t = 0.05$ s
$t = 0.06$ s
$t = 0.07$ s
$t = 0.08$ s